Sri Lanka Institute Of Information Technology



IT3091 - MACHINE LEARNING

PERSONAL LEARNING JOURNEY REPORT

Healthcare – Early Readmission Risk Prediction

Track: Guided Data Track

Group: 2026-DS-21

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Member 1 - Problem Framing & Data Understanding

Name: Rathnayake R.M.A.K.

Student ID: IT24101617

1. My Role in the Project

Our group is developing a machine-learning model to identify hospital encounters that may be associated with readmission within 30 days. We are using the Diabetes 130-US Hospitals dataset, which contains hospital encounter records from 1999 to 2008. My responsibility was to understand the problem, explore the dataset, identify data-quality issues and investigate initial patterns in the data. This helped our team build a clear foundation for the later stages of the project.

2. What I Did

First, I loaded and explored `diabetic_data.csv` and `IDS_mapping.csv` using Python and pandas. I used functions such as `head()`, `info()` and `shape()` to understand the dataset. The original dataset contained 101,766 hospital encounters, 50 columns and 71,518 unique patients.

Next, I defined the unit of analysis as one hospital encounter and helped establish our target variable, `early_readmission`. Encounters labelled `<30` were assigned a value of 1, while encounters labelled `NO` or `>30` were assigned a value of 0.

I then investigated missing values and duplicate records. I found that some variables had substantial missingness, particularly `weight` (96.86%), `max_glu_serum` (94.75%), `A1Cresult` (83.28%) and `medical_specialty` (49.08%). The dataset uses `?` to represent unknown or missing information. Although there were no exact duplicate rows, some patients had multiple encounters.

After applying the project's specified exclusions, 99,340 encounters remained. I also explored readmission patterns according to previous inpatient and emergency visits, length of stay, age and gender. Finally, I identified the potential risk of data leakage caused by repeated records for the same patient. To reduce this risk, the team used patient-grouped data splitting so that encounters from the same patient would not appear in both the training and test sets.

3. Key Findings

One of my main findings was that early readmission was relatively uncommon compared with the other outcomes. After exclusions, 11,314 encounters (11.39%) were classified as early readmissions, while 88,026 (88.61%) belonged to the other class. This imbalance means that accuracy alone may not provide a complete picture of model performance.

I also observed that early readmission rates increased with previous hospital use. The rate rose from approximately 8.6% for patients with no previous inpatient visits to 31.4% for those with four or more. A similar pattern was observed for previous emergency visits: the early readmission rate increased from approximately 10.7% among patients with no previous emergency visits to 25.3% among those with three or more. These results suggest an association between previous hospital use and early readmission, although they do not establish a causal relationship.

Readmission rates also varied across different lengths of hospital stay. However, the rates for female and male patients were quite similar, at approximately 11.46% and 11.30%, respectively. These findings describe patterns in the dataset; they do not prove that any particular factor causes readmission.

4. Challenges and How I Responded

One challenge was dealing with the large number of missing values in some variables. I learned that missing data should be investigated carefully and handled according to the usefulness and reliability of each variable rather than removing all incomplete records automatically.

Another challenge was recognising that the same patient could appear in multiple hospital encounters. I learned why splitting data randomly at the encounter level can cause information leakage and why grouping records by patient is important when evaluating a model.

The imbalanced target variable was another important consideration. I realised that a model could achieve reasonable accuracy by mainly predicting the majority class while failing to identify many early readmissions. Therefore, metrics such as recall, precision, F1-score and, where appropriate, precision-recall AUC can provide a more informative evaluation alongside accuracy.

I also learned to interpret exploratory findings carefully. A relationship between two variables does not necessarily mean that one causes the other.

5. What I Learned

Through this part of the project, I improved my skills in using pandas to inspect datasets, identify missing values, check duplicate records and explore patterns using summary statistics. I also gained a better understanding of target-variable definition, class imbalance and data leakage. Most importantly, I learned that understanding the data before training a model is essential for producing reliable and meaningful results.

6. Reflection and Conclusion

Before working on this project, I mainly thought of data understanding as checking the number of rows and columns and looking for missing values. This experience helped me realise that it involves much more, including defining the problem correctly, understanding what each record represents and deciding how data limitations may affect the model.

My contribution established a clear starting point for the project by defining the prediction target, examining data quality and identifying patterns that deserve further investigation. It also helped the team recognise important challenges before model development. Overall, this experience strengthened my technical skills and taught me to approach machine-learning projects more carefully and critically.